Leyan Li

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Education

University of Edinburgh, United Kingdom — Sep 2025 – Sep 2026
MSc High Performance Computing with Data Science — Expected

University of Liverpool, United Kingdom
BSc Computer Science, Xi'an Jiaotong-Liverpool University 2+2 Programme, First Class Honours — Sep 2023 – Jun 2025

Technical Skills

Programming & Scripting: C/C++, Python, SQL, Bash; automated benchmark execution, log processing, data cleaning, and performance analysis scripts.

HPC & Parallel Computing: MPI, OpenMP, PETSc, Slurm, ARCHER2; MPI-only and hybrid MPI+OpenMP execution, strong scaling, speedup, parallel efficiency, solver benchmarking.

Performance & Systems: Linaro MAP, process/thread models, synchronization overhead, MPI communication overhead, cache locality, NUMA effects, memory bandwidth, file I/O, Linux.

Data, AI & Backend Tooling: Pandas, NumPy, Matplotlib, PyTorch, FastAPI, REST APIs, SQLite/MySQL, Redis, FAISS/Chroma, LangChain/LlamaIndex, Docker, Git.

Projects

HPC Benchmark and Performance Analysis System for PETSc Parallel Solvers | GitHub: https://github.com/lionleepower/petsc-benchmark-suite

Project Summary: Developed a reproducible PETSc benchmark and analysis suite on ARCHER2, comparing MPI-only and hybrid MPI+OpenMP solver performance across 2D/3D stencil problems up to 41.1M unknowns and 128 cores. The workflow automates Slurm submission, PETSc configuration, log ingestion, metric computation, scalability plotting, Linaro MAP profiling, and Markdown/GitHub documentation.

Tech Stack: C/C++, Python, PETSc, MPI, OpenMP, Slurm, Bash, Pandas, NumPy, Matplotlib, Linux, ARCHER2, Linaro MAP

- Engineered an end-to-end benchmark workflow covering compilation, parameter sweeps, Slurm job submission, log collection, CSV normalization, metric calculation, and visualization for reproducible large-scale experiments.
- Executed PETSc linear solver tests on ARCHER2 for 2D/3D stencil benchmarks from 1.0M to 41.1M unknowns, evaluating MPI-only and hybrid MPI+OpenMP configurations up to 128 cores.
- Authored Slurm templates for node count, MPI ranks, OpenMP threads, binding settings, total cores, and PETSc options, including 1-64 node profiling runs with different ranks per node and threads per rank.
- Converted raw Slurm/PETSc outputs into structured experiment records capturing problem size, nodes, ranks, threads, runtime, KSP iterations, solver status, execution mode, and error logs.
- Automated runtime, speedup, and parallel-efficiency calculations with Pandas and Matplotlib, generating scalability plots and comparison tables across problem sizes and core counts.
- Profiled representative large cases with Linaro MAP to separate computation time, MPI communication time, OpenMP region percentage, thread utilization, and potential load imbalance.
- Interpreted efficiency degradation using Amdahl's Law, MPI communication overhead, OpenMP synchronization cost, cache locality, NUMA effects, and memory-bandwidth constraints.
- Documented benchmark commands, experiment configurations, result tables, profiling observations, and optimization notes to make the workflow extensible to additional PETSc examples and matrix formats.

AI Agent System for HPC Benchmark Performance Analysis

Project Summary: Created an 8-layer AI Agent prototype that connects PETSc benchmark logs, structured experiment records, RAG-based HPC documentation retrieval, Redis caching, and Markdown report generation, enabling natural-language analysis of MPI-only and hybrid MPI+OpenMP performance results.

Tech Stack: Python, FastAPI, SQLite/MySQL, Redis, Pandas, Matplotlib, FAISS/Chroma, LangChain/LlamaIndex, Markdown, REST APIs, Linux, Git, Docker

- Architected an 8-layer analysis stack covering API services, log ingestion, metric computation, database storage, RAG retrieval, Agent tool routing, cache management, and observability.
- Delivered FastAPI endpoints for log upload, natural-language queries, historical result retrieval, metric calculation, health checks, and Markdown report generation with validation and error handling.
- Indexed PETSc, MPI, OpenMP, Slurm, Linaro MAP, and personal experiment notes with FAISS/Chroma semantic retrieval to support bottleneck explanations and method lookup.
- Orchestrated tool selection for database queries, log analysis, metric calculation, chart generation, knowledge retrieval, and report assembly based on user questions.
- Added Redis TTL caching and per-request TraceID logging to track cache hits, tool chains, retrieval outputs, database latency, and report-generation latency.

Publication

Li, Leyan. *Convolutional Neural Networks Based Medical Image Analysis.* International Conference on Engineering Management, Information Technology and Intelligence, EMITI 2024.